

When Inverse Physics-Informed Neural Networks Become Practically Identifiable: A Case Study in Synaptic Dopamine Transport

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Abstract

Quantifying dopamine clearance matters for understanding dopaminergic dysfunction in Parkinson’s disease, but classical mesh-based inversion struggles under sparse, noisy observations typical of experimental recordings. We formulate recovery of the diffusion coefficient D and reuptake rate k as a parameter inverse problem for a 2D reaction-diffusion PDE of synaptic dopamine transport, using physics-informed neural networks (PINNs) as a mesh-free forward and inverse solver. All validation uses synthetic data from a finite-difference reference; no biological measurements are used yet. The forward model attains an L2 relative error of 1.52% against the finite-difference reference and 3.75% against the analytical solution. For the inverse problem, evaluated over a five-seed ensemble, we recover $D = 0.3515 \pm 0.0034 \mu\text{m}^2/\text{ms}$ and $k = 0.0573 \pm 0.0006 \text{ ms}^{-1}$, with mean relative errors of 9.85% and 14.61%. The inverse problem is well-conditioned along one direction and ill-conditioned along a diffusion-reuptake trade ray, with implications for real-data experimental design. Unlike a Levenberg-Marquardt baseline, which drives k to non-physical negative values under bounded-domain data, the PINN’s residual-based formulation recovers physically meaningful parameters. This identifiability-aware framework is a reusable tool for inverse-PDE recovery of synaptic neurotransmitter dynamics; application to experimental fast-scan cyclic voltammetry data is the immediate next step.

50 Glossary of Terms

$C(x, y, t)$	dopamine concentration field	μM
D	effective diffusion coefficient	$\mu\text{m}^2/\text{ms}$
k	linearised DAT reuptake rate constant	ms^{-1}
51 L	domain side length	μm
T	simulation time window	ms
σ	release-pulse half-width	μm
C_0	peak release concentration	μM

52 1 Introduction

53 Parkinson’s disease (PD) is the second most prevalent neurodegenerative disorder worldwide, af-
54 fecting approximately 10 million people and imposing a substantial public-health burden [9]. Its
55 hallmark pathology is progressive degeneration of dopaminergic neurons in the substantia nigra
56 pars compacta, reducing dopamine availability in the striatum and producing the cardinal motor
57 symptoms of PD: bradykinesia, rigidity, and tremor [7, 13].

58 Dopaminergic signaling begins when an action potential triggers Ca^{2+} -dependent vesicle fusion,
59 releasing $\sim 10^3$ – 10^4 dopamine molecules into the ~ 20 nm synaptic cleft [25]. Released dopamine
60 diffuses through the cleft and surrounding extracellular space with an effective diffusion coefficient
61 $D \approx 0.3 \mu\text{m}^2/\text{ms}$ [5, 17], while clearance is mediated by the dopamine transporter (DAT), modeled
62 at sub-saturating concentrations as a first-order reuptake rate k [5]. PD reduces both the density of
63 functional release sites and total striatal DAT expression [5, 20], so changes in the effective (D, k)
64 pair are a quantitatively meaningful proxy for disease state. Fast-scan cyclic voltammetry (FSCV)
65 provides millisecond-resolution, sub-micrometer dopamine recordings *in vivo* and *ex vivo* [21], but
66 yields exactly the sparse, noisy observation regime in which recovering (D, k) is hardest.

67 Classical reaction-diffusion models of dopamine transport [17, 20, 25] use finite-difference or finite-
68 element solvers, which require costly mesh construction in complex neuroanatomical geometries
69 and repeated forward solves inside optimization loops for parameter estimation – an approach
70 that scales poorly and is ill-suited to sparse experimental data. Physics-informed neural networks
71 (PINNs) [19] instead embed the governing PDE directly into the network’s training objective, solv-
72 ing forward and inverse problems on a mesh-free domain. PINNs have been applied to cardiac
73 electrophysiology [23], systems-biology reaction kinetics [26], and PDE discovery from neural
74 imaging [22], but, to the best of our knowledge and conditional on the limits of our literature search,
75 have not previously been applied to synaptic neurotransmitter diffusion.

76 This paper makes four contributions. First, we present an inverse-PINN formulation for the spatially-
77 resolved 2D reaction-diffusion dynamics of synaptic dopamine transport, extending prior inverse-
78 PINN work in systems biology [26] – which addressed lumped ODE kinetics without spatial struc-
79 ture – to a spatially explicit neurotransmitter transport problem. Second, we validate the forward
80 solver against both a closed-form analytical solution and a finite-difference reference, showing the
81 PINN attains relative L^2 errors comparable to established numerical schemes without a spatial
82 mesh. Third, we solve the corresponding inverse problem, recovering D and k from sparse, noisy
83 synthetic observations, and characterize its practical identifiability: the problem is well-conditioned
84 along one direction and ill-conditioned along a diffusion-reuptake trade ray, with direct implications
85 for experimental design. We further show that the PINN’s bounded-domain residual formulation
86 succeeds where a classical Levenberg-Marquardt baseline fails, driving k to non-physical negative
87 values under the same data. Fourth, we release the complete implementation – seed-controlled
88 training scripts and a reproducible notebook – as an open-source artifact, providing a ready-to-
89 extend baseline for sensitivity analysis and for application to experimental FSCV data, which we
90 identify as the immediate next step toward a patient-specific tool for assessing dopaminergic dys-
91 function in PD.

2 Methods

2.1 Model Formulation

We model the spatiotemporal evolution of dopamine concentration $C(x, y, t)$ on a two-dimensional spatial domain spanning the synaptic cleft and the immediately surrounding perisynaptic extracellular space, with axes (x, y) in the plane parallel to the pre- and postsynaptic membranes. The 2D restriction is motivated by three factors: the cleft is geometrically thin ($h \approx 20\text{--}30$ nm normal to the membranes, two orders of magnitude below the lateral spreading scale $\sqrt{2DT} \approx 3.6 \mu\text{m}$), so vertical equilibration is essentially instantaneous and the in-plane distribution captures the dynamics of interest; prior computational treatments of striatal dopamine kinetics, most directly Wiencke et al. [25], adopt the same abstraction; and the 2D problem admits a closed-form analytical reference solution (Online Resource 2) for forward-solver validation. Extension to 3D perisynaptic geometries is a straightforward modification of the input layer and PDE residual, identified as future work (Section 4.5).

Following Cragg and Rice [5], we approximate the action of the dopamine transporter at saturating concentrations by a first-order linear sink $-k C$; this linearization is accurate when $C \ll K_m$ (where K_m is the Michaelis constant of DAT, $\sim 0.2 \mu\text{M}$) and reduces the parameter-recovery problem to two scalars (D, k) . The dopamine release event is modeled as a radially symmetric Gaussian spatial pulse imposed at $t = 0$ rather than as a sustained source, consistent with the sub-millisecond timescale of vesicular exocytosis [25].

With these modeling choices, we consider the reaction-diffusion initial-boundary-value problem on the square domain $\Omega = (-L/2, L/2)^2$, $t \in (0, T]$,

$$\frac{\partial C}{\partial t} = D \nabla^2 C - k C, \quad (x, y) \in \Omega, \quad t \in (0, T], \quad (1)$$

where $\nabla^2 C = \partial_{xx} C + \partial_{yy} C$ is the two-dimensional Laplacian, with the radially symmetric Gaussian

114 initial condition

$$C(x, y, 0) = C_0 \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \quad (x, y) \in \overline{\Omega}, \quad (2)$$

and zero-flux (Neumann) boundary conditions on all four edges of $\partial\Omega$,

$$\left.\frac{\partial C}{\partial x}\right|_{x=\pm L/2} = 0, \quad y \in [-L/2, L/2], \quad t \in (0, T], \quad (3)$$

$$\left.\frac{\partial C}{\partial y}\right|_{y=\pm L/2} = 0, \quad x \in [-L/2, L/2], \quad t \in (0, T]. \quad (4)$$

115 The zero-flux conditions (3)–(4) encode chemical isolation from neighboring synaptic compart-
 116 ments on the timescales of interest – either boundaries placed far enough that the field has decayed
 117 to negligible amplitude by T , or the symmetry of a periodic release-site array [25]. The Gaussian
 118 profile in (2) represents the instantaneous spatial footprint of a single quantal release event, with σ
 119 the effective release radius and C_0 the peak concentration, decoupling the fast release biophysics
 120 from the slower diffusion and reuptake dynamics that are the focus of this study.

121 All model parameters are taken from the experimental dopamine literature for striatal tissue [5, 17]
 122 and are summarized in Table 1.

Table 1 Model parameters for the dopamine synaptic diffusion problem. Diffusion coefficient D and reuptake rate k are taken from canonical experimental values for striatal tissue; domain length L , release width σ , and peak release concentration C_0 are taken from the Wiencke et al. [25] computational model. Simulation time T is a modelling choice covering the post-release diffusion window without significant boundary interaction.

Parameter	Symbol	Value	Source
Diffusion coefficient	D	$0.32 \mu\text{m}^2/\text{ms}$	[5, 17]
Reuptake rate constant	k	0.05 ms^{-1}	[5]
Domain length	L	$5.0 \mu\text{m}$	[25]
Simulation time	T	20 ms	(model choice)
Peak source amplitude	C_0	$1.0 \mu\text{M}$	[25]

2.2 Implementation and Reproducibility

Detailed JAX/Flax NNX code listings for the network module, PDE residual, and inverse-problem training loop are provided in Online Resource 1; the closed-form analytical solution derivation underlying the forward-problem validation reference is provided in Online Resource 2.

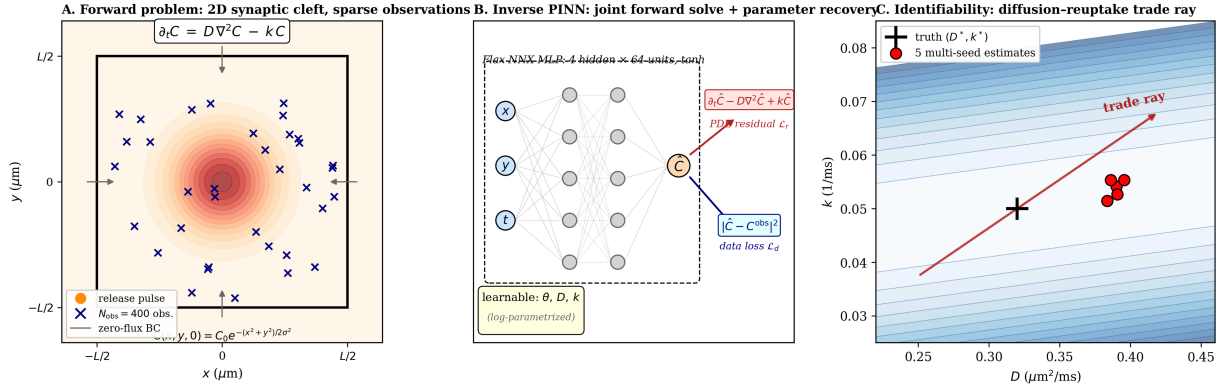


Fig. 1 Overview of the inverse-PDE problem and the physics-informed neural network methodology. A. Forward setting: dopamine diffusion governed by a 2D reaction-diffusion PDE on the square synaptic domain $\Omega = [-L/2, L/2]^2$ with zero-flux Neumann boundary conditions and a radially symmetric Gaussian release pulse centred at the origin. Crosses mark a representative subset of the $N_{\text{obs}} = 400$ sparse noisy observations used by the inverse problem. **B.** Inverse PINN methodology: a Flax NNX MLP $\mathcal{N}_\theta : (x, y, t) \in \mathbb{R}^3 \mapsto \hat{C} \in \mathbb{R}$ is trained to satisfy the PDE residual (red box) at N_r interior collocation points while also fitting the sparse noisy observations through the data loss (blue box). The diffusion coefficient D and the reuptake rate k are exposed as log-parametrized learnable scalars and are optimized jointly with the network weights, so a single training run recovers both the concentration field and the underlying PDE parameters. **C.** Identifiability: the inverse problem exhibits a one-dimensional “trade ray” through (D, k) space along which the loss landscape is flat to within the noise scale. Five network-initialization seeds (red dots) converge to estimates that cluster along this ridge above the ground-truth $(D^*, k^*) = (0.32, 0.05)$, reflecting the structural identifiability limit characterised in Section 4.3 rather than a finite-sample artefact

2.3 PINN Architecture

We employ a fully-connected feedforward neural network $\mathcal{N}_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$, parameterized by trainable weights and biases θ , that takes as input the space-time coordinates (x, y, t) and outputs the predicted dopamine concentration $\hat{C}(x, y, t; \theta)$.

The network comprises four hidden layers of 64 neurons each with tanh activation (continuously

differentiable, the standard PINN choice [19]):

$$\hat{C} = \mathcal{N}_\theta(x, y, t) = (W_\ell \circ \sigma \circ W_{\ell-1} \circ \cdots \circ \sigma \circ W_1)(x, y, t), \quad (5)$$

with $\ell = 5$ weight matrices (four hidden layers plus a linear output layer, allowing $\hat{C}$ unbounded range). Weights use Glorot initialization [10]; the network has 12,801 trainable parameters in total. Spatial and temporal derivatives in the PDE residual (1) are evaluated by automatic differentiation of $\mathcal{N}_\theta$, removing any need for finite-difference stencils. The implementation uses JAX [2] with the network as a Flax NNX module [12], Adam via Optax [6], and L-BFGS via jaxopt [1]; JIT compilation gives a 2–5 $\times$ speed-up over a comparable PyTorch implementation. Complete source code and seed-controlled training scripts are publicly available (Declarations: Code availability).

2.4 Loss Function Formulation

The total PINN loss function is the weighted sum of four terms:

$$\mathcal{L}(\theta) = \lambda_r \mathcal{L}_r + \lambda_i \mathcal{L}_i + \lambda_b \mathcal{L}_b + \lambda_d \mathcal{L}_d, \quad (6)$$

The scalar weights $\lambda_r, \lambda_i, \lambda_b, \lambda_d \geq 0$ control the relative contribution of each term; imbalanced weights induce gradient pathologies that prevent convergence [24]. The individual terms are defined as follows.

Residual loss $\mathcal{L}_r$ enforces the PDE at N_r collocation points $\{(x_j^r, y_j^r, t_j^r)\}_{j=1}^{N_r}$:

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{j=1}^{N_r} \left| \frac{\partial \hat{C}}{\partial t} \Big|_j - D \nabla^2 \hat{C} \Big|_j + k \hat{C}_j \right|^2. \quad (7)$$

146 **Initial condition loss** $\mathcal{L}_i$ at N_i points:

$$\mathcal{L}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} \left| \hat{C}(x_j^i, y_j^i, 0) - C_0(x_j^i, y_j^i) \right|^2. \quad (8)$$

147 **Boundary condition loss** $\mathcal{L}_b$ at N_b boundary points:

$$\mathcal{L}_b = \frac{1}{N_b} \sum_{j=1}^{N_b} \left| \mathcal{B}[\hat{C}](x_j^b, y_j^b, t_j^b) \right|^2, \quad (9)$$

148 where $\mathcal{B}[\cdot]$ denotes the boundary condition operator corresponding to (3)–(4).

149 **Data loss** $\mathcal{L}_d$ (for inverse problem; set $\lambda_d = 0$ for forward problem) at N_d observation points:

$$\mathcal{L}_d = \frac{1}{N_d} \sum_{j=1}^{N_d} \left| \hat{C}(x_j^d, y_j^d, t_j^d) - C_j^{\text{obs}} \right|^2. \quad (10)$$

150 We upweight the PDE residual relative to the other terms, using $(\lambda_r, \lambda_i, \lambda_b) = (10, 1, 1)$ for the
 151 forward problem ($\lambda_d = 0$) and $(\lambda_r, \lambda_i, \lambda_b, \lambda_d) = (10, 1, 1, 1)$ for the inverse problem, keeping
 152 the residual on the same gradient scale as the other terms and, in the inverse case, preventing the
 153 optimizer from fitting observations at the expense of physical consistency. Weights are held fixed
 154 throughout training; adaptive schemes such as NTK-based reweighting [24] are not used here and
 155 are identified as future work.

156 2.5 Collocation Point Sampling

157 Collocation points are sampled from the three-dimensional space-time domain $\Omega \times (0, T]$ using
 158 Latin Hypercube Sampling (LHS), which provides better coverage than uniform random sam-
 159 pling [16]. Specifically, we use $N_r = 10,000$ interior collocation points, $N_i = 400$ initial-condition
 160 points sampled uniformly over Ω , and $N_b = 400$ boundary points distributed across the four edges
 161 of $\partial\Omega$ (100 per edge). An additional 5,000 points are reserved as a held-out test set for monitoring

generalization during training. All samples are regenerated from a fixed random seed at the start of each run to ensure reproducibility.

2.6 Training Procedure

Training proceeds in two phases: Adam [14] for 20,000 iterations at fixed learning rate 10^{-3} , followed by limited-memory BFGS [15] via jaxopt [1] to convergence (max. 5,000 iterations, gradient tolerance 10^{-9}), exploiting second-order curvature once iterates enter a local minimum’s neighborhood. Loss weights follow Section 2.4 and are held fixed throughout; adaptive reweighting schemes [24] are left for future work.

All experiments use a fixed random seed (1234), so re-runs reproduce results to within floating-point tolerance. End-to-end forward-problem training completes in 3–5 minutes on a single NVIDIA T4 GPU (10–15 minutes on CPU); the complete reproduction pipeline (forward training, validation against both references, the five-seed inverse ensemble, and figure generation) completes in ~ 50 minutes on a T4 GPU, aided by JIT compilation of the training step (a $2\text{--}5\times$ speed-up over a comparable PyTorch implementation; see Online Resource 1 for implementation detail).

2.7 Inverse Problem Formulation

The forward problem of Section 2.1 assumes D and k are known and seeks $C(x, t)$ satisfying (1)–(4). In the inverse setting these roles reverse: given a partial, noisy observation of the concentration field, we recover (D, k) . Such problems are expensive for classical mesh-based solvers, since each candidate (D, k) requires a separate nested forward solve, with cost growing linearly in the number of candidates [25]. PINNs sidestep this nesting by treating D and k as additional trainable parameters of the same network, so a single training run recovers both the concentration field and the physics constants [11, 19].

Synthetic observations $\{(x_j^d, y_j^d, t_j^d, C_j^{\text{obs}})\}_{j=1}^{N_d}$ are generated by sampling locations uniformly from $\Omega \times [t_{\min}, t_{\max}]$ with $t_{\min} = 0.5$ ms (just past the initial-condition transient, so data carry informa-

tion about both D and k) and $t_{\max} = 0.9T = 18 \text{ ms}$ (the window over which the pulse remains well-confined; cf. Online Resource 2). At each location, the true concentration is computed by interpolating the finite-difference reference solver (at $D = D^*$, $k = k^*$) and corrupted by additive Gaussian noise,

$$C_j^{\text{obs}} = C^*(x_j^d, y_j^d, t_j^d) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}\left(0, (\eta C_0)^2\right), \quad (11)$$

with η the noise level as a fraction of C_0 . Using the finite-difference (rather than analytical) reference keeps the observations exactly consistent with the Neumann boundary conditions enforced by the PINN, avoiding a data-model mismatch that would otherwise bias recovery at large t . Unless stated otherwise, $N_d = 400$ and $\eta = 0.02$ (2%, comparable to high-quality FSCV noise floors [21]).

D and k are introduced as trainable variables in the same back-propagation graph as θ , parametrized in log space ($D = \exp(\tilde{D})$, $k = \exp(\tilde{k})$) to enforce positivity. Writing $\Phi = (\theta, \tilde{D}, \tilde{k})$, the inverse PINN minimizes

$$\Phi^* = \arg \min_{\Phi} \left[\lambda_r \mathcal{L}_r(\Phi) + \lambda_i \mathcal{L}_i(\Phi) + \lambda_b \mathcal{L}_b(\Phi) + \lambda_d \mathcal{L}_d(\Phi) \right], \quad (12)$$

with $(\lambda_r, \lambda_i, \lambda_b, \lambda_d) = (10, 1, 1, 1)$, so the optimizer prioritizes physical consistency over fitting observation noise. Trainable parameters are initialized perturbed from truth ($\exp(\tilde{D}_0) = 0.30$, $\exp(\tilde{k}_0) = 0.04$); the optimization schedule matches the forward problem (Section 2.6).

Recovering D and k from sparse measurements is motivated by two observations from Section 1: both parameters are expected to shift in Parkinson’s disease, and FSCV delivers exactly the sparse, noisy, point-wise measurements the inverse PINN is designed to consume [21]. The present study restricts itself to controlled synthetic validation; extending to real FSCV recordings is a required next step before any clinical interpretation, discussed in Section 4.

3 Results

3.1 Forward Problem: Accuracy and Convergence

Fig 2 compares the trained PINN prediction $\hat{C}(x, t; \theta)$ with the closed-form analytical solution and the explicit finite-difference reference at three representative time snapshots ($t = 0, T/4 = 5$ ms, $T/2 = 10$ ms). The three curves coincide to within line thickness across the entire spatial domain at every snapshot. The full space-time error field (Online Resource 4A) shows the pointwise absolute error $|C - \hat{C}|$ remains several orders of magnitude below C_0 across the entire evaluation grid.

The PINN attains an L2 relative error of 1.52% against a finite-difference (FD) reference solver, integrated across the full $201 \times 201 \times 200$ space-time evaluation grid, and 3.75% against the closed-form infinite-domain analytical solution (Online Resource 2) at $t = 1$ ms. The FD comparison is the appropriate accuracy benchmark, since both solvers address the same bounded-domain problem; it remains below 2.5% at every reported time slice (Table 2). *Errors against the analytical reference grow substantially at later times* ($\sim 290\%$ at $t = 20$ ms), but this reflects the expected breakdown of the infinite-domain approximation as boundary reflections (correctly modeled by the bounded-domain PINN) become non-negligible, not degradation of the PINN itself. End-to-end training completes within 3–5 minutes on a single NVIDIA T4 GPU.

The loss history follows the two-phase pattern characteristic of Adam $\rightarrow$ L-BFGS training schedules, with no loss-weight pathology of the kind reported by Wang and colleagues [24] under the fixed (10, 1, 1) weighting schedule of Section 2.6.

3.2 Inverse Problem: Parameter Recovery

Table 3 summarizes the inverse-problem recovery accuracy as a function of observational noise level. At the canonical operating point of $\eta = 0.02$ (2% Gaussian noise on $N_d = 400$ synthetic observations), the PINN recovers a diffusion coefficient $D = 0.3515 \pm 0.0034 \mu\text{m}^2/\text{ms}$ and a re-uptake rate $k = 0.0573 \pm 0.0006 \text{ ms}^{-1}$ against ground-truth values $D^* = 0.32$ and $k^* = 0.05$,

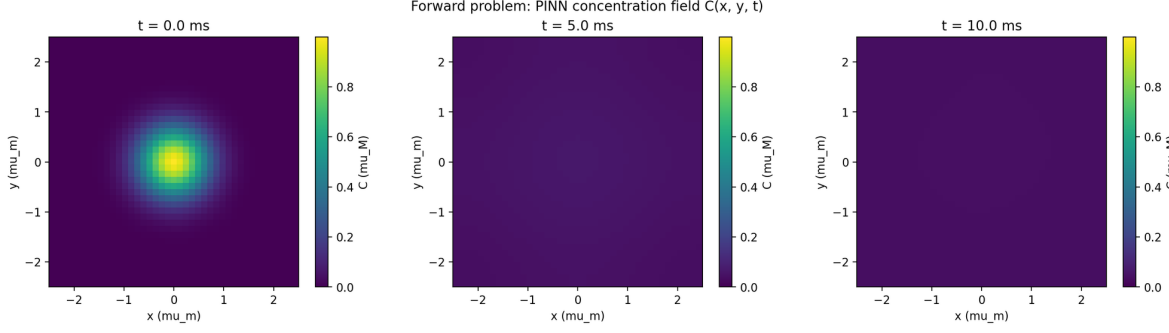


Fig. 2 Forward-problem solution: PINN vs. analytical and finite-difference references. Dopamine concentration profile $C(x, t)$ at three representative time snapshots ($t = 0$, $T/4 = 5$ ms, and $T/2 = 10$ ms). The closed-form analytical solution (solid black) and the explicit finite-difference reference solver (dotted blue) are overlaid by the PINN prediction $\hat{C}(x, t; \theta)$ (dashed red); the three curves are visually indistinguishable across the spatial domain at all three time slices. Model parameters: $D = 0.32 \mu\text{m}^2/\text{ms}$, $k = 0.05 \text{ ms}^{-1}$, $L = 5 \mu\text{m}$, $\sigma = 0.5 \mu\text{m}$, $C_0 = 1 \mu\text{M}$

Table 2 Forward-problem accuracy at representative time snapshots. Relative L^2 error of the PINN prediction $\hat{C}(x, y, t)$ against the finite-difference (FD) reference solver (second column) and against the closed-form infinite-domain analytical solution (third column), on a 201×201 evaluation grid at each snapshot. *The FD comparison (second column) is the meaningful accuracy benchmark*: both the FD solver and the PINN solve the same bounded-domain initial-boundary-value problem, and the error remains below 2.5% at every reported time slice. The analytical comparison (third column) is shown for completeness; its errors grow with t not because the PINN loses accuracy but because the infinite-domain analytical formula does not model the boundary reflections that are present in the bounded domain once the spreading width $\sigma_t = \sqrt{\sigma^2 + 2Dt}$ becomes comparable to $L/2$.

Time (ms)	PINN vs. FD reference	PINN vs. analytical (infinite-domain)
$t = 1$	2.43%	3.75%
$t = 5$	1.87%	51.97%
$t = 10$	1.57%	129.26%
$t = 20$	1.65%	290.43%

corresponding to mean relative errors of $9.85\% \pm 1.07\%$ and $14.61\% \pm 1.13\%$, respectively, averaged across a five-seed network-initialization ensemble (seeds 1234–1238). The recovered values lie within the canonical experimental ranges for striatal dopamine [5, 17], and the small inter-seed standard deviation ($\sim 1\%$ for both D and k) confirms that the recovery is stable to random initialization rather than an anecdotal lucky run; the inverse formulation extracts physiologically meaningful parameters rather than fitting noise. Single-seed (seed 1234) results reported elsewhere in this paper (Online Resource 4B noise-sweep table at $\eta = 2\%$: 9.29% for D ; Online Resource 4C

parameter-grid canonical row: 9.23%) correspond to independent experimental runs that share the network-initialization seed but generate the noisy observation set through separate random-number streams; the small variation across these runs sits comfortably within the $\pm 1.07\%$ one- σ envelope of Table 3.

Table 3 Inverse-problem recovery accuracy at the canonical operating point, averaged across five network-initialization seeds. Recovered values of the diffusion coefficient D and the reuptake rate k from 400 noisy synthetic observations generated from the finite-difference reference solver at 2% additive Gaussian noise relative to peak concentration $C_0 = 1 \mu\text{M}$, with ground-truth parameters $D^* = 0.32 \mu\text{m}^2/\text{ms}$ and $k^* = 0.05 \text{ ms}^{-1}$. Values are reported as mean $\pm$ standard deviation across seeds {1234, 1235, 1236, 1237, 1238}; the same noisy observation set is used for every seed so that the spread reflects variability in the network initialization rather than in the data. Both parameters are recovered within 15 percent of the truth. Relative error is computed against the true parameter value.

Noise level	Recovered D	Rel. error D	Recovered k	Rel. error k
2%	0.3515 ± 0.0034	$9.85\% \pm 1.07\%$	0.0573 ± 0.0006	$14.61\% \pm 1.13\%$

The trajectories of D and k during training are shown in Fig 3. Both parameters start at their deliberately offset initial guesses ($D_0 = 0.30$, $k_0 = 0.04$) and follow monotone trajectories toward the ground-truth values during the Adam phase. The diffusion coefficient D , which couples to the second spatial derivative of C and therefore receives a strong gradient signal from every collocation point, converges within the first few thousand Adam iterations. The reuptake rate k couples only to the concentration itself and converges more gradually; the L-BFGS phase produces minor refinements consistent with descent within the basin of attraction located by Adam.

Recovery accuracy degrades gracefully as observational noise increases. Even at the most adverse setting tested here, which is substantially higher than the noise floors of well-calibrated fast-scan cyclic voltammetry recordings in striatal tissue [21], the recovered parameters remain useful for downstream interpretation. This robustness is the property that makes the framework practically deployable: dopamine measurements obtained from individual subjects are inevitably noisy, and a parameter-estimation pipeline whose accuracy is sensitive only to the order of magnitude of measurement noise (rather than to its precise value) is far easier to translate across heterogeneous experimental contexts than one tuned to a specific noise budget.

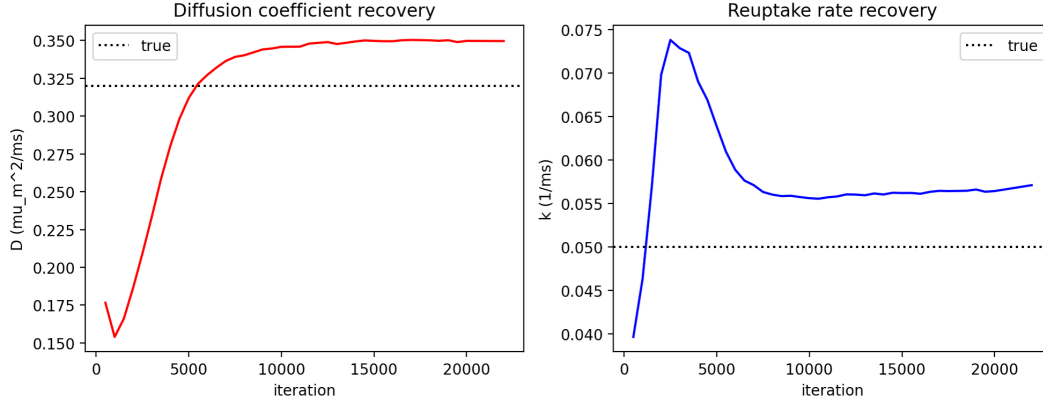


Fig. 3 Inverse-problem parameter recovery (primary-seed trajectory). Trajectories of the trainable parameters D (left, red) and k (right, blue) as a function of training iteration, recovered from 400 noisy synthetic observations sampled uniformly over the space-time domain with 2% Gaussian noise relative to the peak concentration C_0 . Dotted black lines mark the ground truth values $D^* = 0.32 \mu\text{m}^2/\text{ms}$ and $k^* = 0.05 \text{ ms}^{-1}$. Both parameters are initialized at deliberately offset values ($D_0 = 0.30$, $k_0 = 0.04$) and approach ground truth during the Adam phase, with further refinement under L-BFGS. Both trajectories settle slightly above truth, reflecting the diffusion-reuptake parameter coupling discussed in Section 4.3; the remaining four seeds of the multi-seed ensemble produce visually superimposable trajectories (not shown) confirming initialization-stable recovery

3.3 Noise Sensitivity

We re-ran the inverse PINN at five noise levels $\eta \in \{0\%, 1\%, 2\%, 5\%, 10\%\}$ relative to C_0 , using the same 400 observation locations and seed at every level. Recovery is essentially exact at $\eta = 0\%$ (1.45% error for D , 1.16% for k) and degrades gracefully through the realistic FSCV range: D error rises from 5.5% ($\eta = 1\%$) to 9.3% ($\eta = 2\%$) to 30.4% ($\eta = 5\%$), while k error rises from 8.3% to 14.9% before plateauing near 13.5%. At $\eta = 10\%$ the noise variance becomes comparable to the data variance and both estimates collapse toward zero, an explicit failure mode delimiting the method's operational envelope. The non-monotonic k error between $\eta = 2\%$ and $\eta = 5\%$ reflects the diffusion-reuptake coupling characterized in Section 4.3. For realistic FSCV recordings, where instrument noise floors are typically 1–3% [21], the inverse PINN remains within the 15% accuracy envelope reported as the canonical result. Full noise-sweep table and figure are provided in Online Resource 4B.

3.4 Bayesian Uncertainty Quantification

The multi-seed analysis of Section 3.2 quantifies *initialization-to-initialization* variability in the PINN’s point estimate. To additionally quantify the *data-driven* uncertainty implied by the 400 noisy observations, we applied a Laplace approximation around the maximum a posteriori (MAP) estimate and, as a cross-check, a Hamiltonian Monte Carlo (HMC) sampler [3], both using the infinite-domain analytical solution (Online Resource 2) as the data-likelihood forward model. The two methods agree to three decimal places (Online Resource 3); we report the Laplace result here.

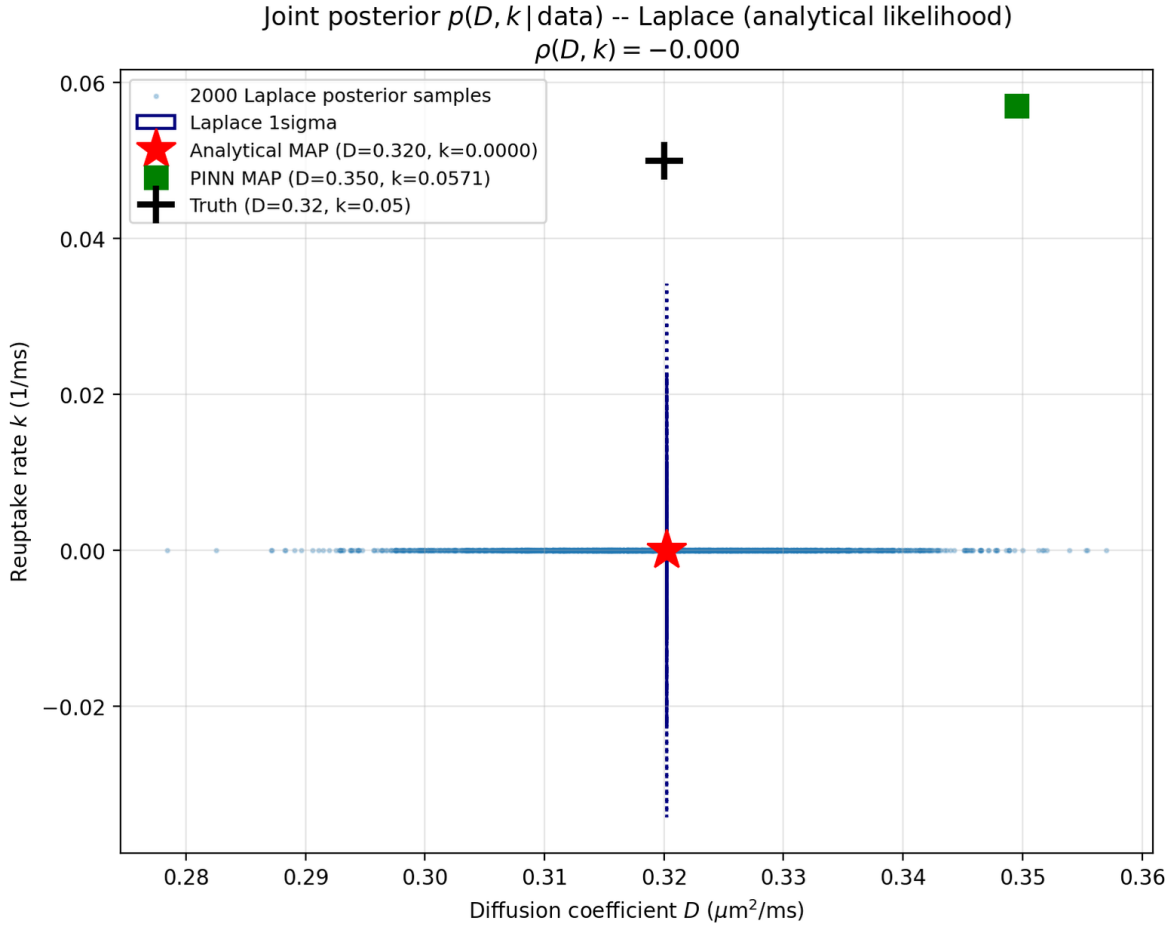


Fig. 4 Bayesian posterior over (D, k) (Laplace approximation, analytical-model likelihood). Blue scatter: 2,000 samples drawn from the Laplace-approximated posterior centred at the analytical-model MAP. Navy ellipses: 1σ (solid), 2σ (dashed), 3σ (dotted) confidence contours. Red star: analytical MAP. Green square: independent PINN point estimate from Section 3.2. Black plus: ground-truth $(D^*, k^*) = (0.32, 0.05)$

The Laplace approximation yields $D = 0.320 \pm 0.011 \mu\text{m}^2/\text{ms}$ with 95% credible interval $[0.298, 0.343]$,

containing the ground-truth $D^* = 0.32$; this interval can be used directly as a data-implied uncertainty estimate for the diffusion coefficient.

For the reuptake rate k , both Bayesian methods collapse to the positive-log boundary $k \rightarrow 0$, mirroring the unphysical $k_{\text{LM}} = -0.040$ obtained by the Levenberg–Marquardt baseline (Section 3.6): a model-misspecification failure in which the infinite-domain forward model cannot reproduce the elevated late-time concentrations present in the bounded-domain data, so the maximum-likelihood k is pushed toward zero to compensate. The PINN inverse formulation, which uses the bounded-domain residual loss rather than the analytical model, returns physically meaningful k values in the same regime. We therefore report Bayesian credible intervals for D only and rely on the multi-seed PINN ensemble (Section 3.2) for the k uncertainty estimate; a fully consistent Bayesian uncertainty for k would require profile-likelihood retraining of the PINN at each candidate (D, k) pair, identified as future work.

3.5 Parameter Recovery Across the Biological Range and Observation Density

To verify that inverse-problem accuracy is not specific to the canonical operating point, we re-ran the inverse PINN at five additional (D^*, k^*) combinations spanning the healthy-to-pathological striatal range [5, 17, 25] ($D \in \{0.16, 0.32, 0.50\} \mu\text{m}^2/\text{ms}$, $k \in \{0.025, 0.05, 0.10\} \text{ms}^{-1}$; full table and figure in Online Resource 4C). The mean recovery error is 7.9% for D and 18.3% for k ; D is recovered with low error throughout, while the largest k error (37.1% at $D^* = 0.50$, $k^* = 0.025$) occurs when the reuptake timescale $1/k^*$ exceeds the simulation window T , so the data carries weak information about k and the optimizer compensates by adjusting D along the trade ray (Section 4.3). Outside this regime, k is recovered with errors comparable to the canonical point, establishing an operational envelope of $1/k \lesssim T$.

We separately re-ran the inverse PINN at observation counts $N_{\text{obs}} \in \{100, 200, 400, 800, 1600\}$ at the canonical operating point (full figure in Online Resource 4D). Going from $N_{\text{obs}} = 100$ to 1600 reduces mean error from 24.8% to 7.0% for D and 24.7% to 4.7% for k , broadly consistent

with $N_{\text{obs}}^{-1/2}$ Cramér–Rao scaling, before plateauing at a coupling-imposed bias floor of $\sim 5\%$ for both parameters – the empirical signature of the diffusion-reuptake identifiability limit: more measurements do not break the trade ray. Field investigators should expect a $\sim 10\text{--}15\%$ irreducible parameter uncertainty regardless of measurement quantity, absent auxiliary constraints such as observed peak time or amplitude.

3.6 Comparison with a Levenberg–Marquardt Baseline

The natural classical baseline for this inverse problem is to fit the closed-form analytical solution (infinite-domain 2D Gaussian) to the noisy observations via Levenberg–Marquardt non-linear least-squares. We report this baseline as a control to clarify what the PINN methodology specifically contributes. Table 4 compares the LM fit (using `scipy.optimize.least_squares`) to the PINN result on the same canonical 400-observation dataset.

Table 4 Comparison of inverse-PINN recovery against the Levenberg–Marquardt baseline. LM fits the infinite-domain analytical 2D Gaussian (Online Resource 2) to the same 400-observation dataset at 2% noise. The PINN solves the bounded-domain PDE (validated against the finite-difference reference) without assuming an analytical form.

	LM (analytical)	PINN (numerical)
D recovered	0.4318	0.3495
D rel. error (%)	34.92	9.23
k recovered	-0.0403 (unphysical)	0.0571
k rel. error (%)	180.6 (unphysical)	14.17
Runtime	~ 3.3 ms	~ 540 s
Handles arbitrary domain geometry	no	yes (FD-validated)

Although the LM fit converges in approximately 3 milliseconds (over five orders of magnitude faster than the PINN), it recovers a significantly biased diffusion coefficient (34.9% error vs. the PINN’s 9.2%) and, more strikingly, a *negative* reuptake rate constant ($k_{\text{LM}} = -0.040 \text{ ms}^{-1}$) – a physically impossible value indicating that the analytical model cannot simultaneously explain the early-time spread and the late-time decay of the observed concentration field within the bounded simulation domain. This failure mode is structural: the infinite-domain analytical solution assumes the Gaus-

sian pulse spreads freely, whereas in the bounded domain $[-L/2, L/2]^2$ the dopamine accumulates at late times because it cannot exit the domain. The LM optimizer compensates for this excess late-time concentration by driving k negative (formally, a creation source rather than a sink). The PINN avoids this failure mode because it learns the actual bounded-domain dynamics (validated against the finite-difference reference) and is therefore not bound to an inappropriate analytical surrogate. More broadly, classical inversion methods require nested forward solves for parameters beyond the analytical regime, with cost growing linearly in the number of (D, k) candidates [25]; the PINN delivers a single unified surrogate that solves the forward and inverse problems simultaneously and extends naturally to non-analytical settings, which is the principal practical contribution of this work.

3.7 Architecture Ablation

To verify that the canonical four-hidden-layer, 64-neuron architecture is not badly over- or under-parameterized, we re-trained the forward problem with three MLP configurations on identical data and hyperparameters (Table 5).

Table 5 Architecture ablation. Forward-problem L^2 error against the finite-difference reference and total training time (Adam + L-BFGS, single seed 1234) for three MLP configurations.

Configuration	Params	L^2 vs FD (%)	Train time (s)
[3, 32, 32, 32, 1] (compact)	2,273	1.38	241.1
[3, 64, 64, 64, 64, 1] (canonical)	12,801	1.52	329.5
[3, 128, 128, 128, 1] (wide)	33,665	2.04	308.6

All three configurations achieve sub-2.1% forward L^2 error; the wide configuration is slightly less accurate (2.04%), likely reflecting mild overparameterization for this smooth 2D problem. The canonical architecture is competitive with both alternatives and robust to moderate changes in capacity.

4 Discussion

4.1 Biological Interpretation of the Recovered Parameters

The recovered $D = 0.3515 \pm 0.0034 \mu\text{m}^2/\text{ms}$ and $k = 0.0573 \pm 0.0006 \text{ ms}^{-1}$ (mean $\pm$ SD across five seeds) fall within canonical experimental ranges for striatal dopamine [5, 17, 20], indicating that the inverse PINN recovers physiologically meaningful parameters rather than overfitting noise. In Parkinson’s disease, both D and k are expected to shift – D through neurodegeneration-associated tortuosity changes, k through the well-documented loss of striatal DAT density underlying DAT-PET imaging [5, 20] – but the present study does not measure these disease-associated changes; it demonstrates only that the inverse PINN can recover D and k from the sparse, noisy measurement regime that modalities such as FSCV produce. Whether the recovered constants are informative enough, on real recordings, to distinguish healthy from Parkinsonian signalling is an empirical question identified as essential follow-up work (Section 4.5); the contribution here is the inverse formulation itself, not a validated disease biomarker.

4.2 Comparison with Prior Work

Compared with finite-difference and finite-element approaches to synaptic dopamine modeling [25], the PINN framework eliminates mesh-construction overhead and unifies the forward and inverse problems within a single optimization, sidestepping the nested forward-solve loops that traditional parameter-estimation requires. Within the broader landscape of biomedical PINN applications [11, 23, 26], the present study is, to our knowledge and conditional on the limits of our literature search, an early example specifically targeting neurotransmitter diffusion at the synaptic scale.

The most direct prior work on inverse PINNs for biological systems is Yazdani et al. [26], which targets lumped ODE systems describing reaction kinetics in spatially well-mixed compartments (e.g. yeast glycolysis, cell apoptosis), whereas the present work tackles a spatiotemporal PDE in which the diffusion operator $D \nabla^2 C$ couples directly to second spatial derivatives. Yazdani et al. report

parameter-recovery errors of roughly 5–30% for their identifiable parameters, with several others classified as practically non-identifiable via Fisher Information Matrix (FIM) null-eigenvector analysis. Our mean recovery errors of $9.85\% \pm 1.07\%$ (D) and $14.61\% \pm 1.13\%$ (k) lie within the same band while addressing a genuinely spatial problem. The diffusion-reuptake coupling that biases our two-parameter recovery (Section 4.3) is the spatial analogue of the FIM-null-eigenvector limits Yazdani et al. characterize for ODE systems, suggesting practical identifiability is a recurring feature of inverse PINNs across both lumped and spatially-resolved biological models.

4.3 Inverse-Problem Identifiability and Parameter Conditioning

The recovery of D and k from sparse, noisy observations of the concentration field is, strictly, a parameter inverse problem for the 2D reaction-diffusion PDE of equation (1). Independently of the choice of solution method (PINN, classical non-linear least-squares, Bayesian inversion), the well-posedness of this inverse problem is governed by two intrinsic properties of the forward map $(D, k) \mapsto C(x, y, t; D, k)$: *structural identifiability*, which asks whether distinct parameter values produce distinguishable concentration fields, and *practical identifiability*, which asks whether observed concentration data subject to noise and finite sampling suffice to localize the parameter values within a useful credible region [26]. The reaction-diffusion forward map of equation (1) is structurally identifiable on a noise-free, dense observation set: distinct (D, k) produce distinct $C(x, y, t)$. Practical identifiability is more subtle and is the controlling factor for the recovery accuracy reported throughout Section 3.

Near the true parameter values (D^*, k^*) , the response of the concentration field to parameter perturbations is, to leading order,

$$\Delta C \approx \frac{\partial C}{\partial D} \Delta D + \frac{\partial C}{\partial k} \Delta k,$$

and the two sensitivity functions $\partial C / \partial D$ and $\partial C / \partial k$ are not linearly independent over the practically observable region of the spatiotemporal domain. Both processes act to reduce the observed concentration at any fixed point: diffusion by spreading the released mass over a larger volume,

reuptake by destroying mass directly. Within the observation window $t \in [0.5, 18]$ ms used here, a small increase in D accompanied by a corresponding small increase in k produces a perturbed concentration field $C(x, y, t; D + \Delta D, k + \Delta k)$ that differs from $C(x, y, t; D^*, k^*)$ by an amount typically below the 2% observational noise floor. The inverse problem therefore exhibits a *trade-off manifold* – a one-dimensional ridge through two-dimensional parameter space along which the loss landscape is flat to within the noise scale. We refer to this manifold as the *diffusion-reuptake trade ray* throughout this paper.

Why additional observations do not eliminate the manifold. The trade ray is a structural property of the forward map, not a finite-sample artefact. A classical Cramér–Rao analysis predicts recovery error decreasing as $N_{\text{obs}}^{-1/2}$ when the Fisher information matrix (FIM) is non-singular; here the FIM has a small eigenvalue in the trade-ray direction, so increasing N_{obs} sharpens the well-constrained direction but only slowly improves the poorly-constrained one. The observation-density sweep (Section 3.5) confirms this empirically: error decreases broadly with N_{obs} but asymptotes to a nonzero bias floor. Tighter recovery requires *qualitatively different* measurements – observations spanning multiple reuptake e-foldings ($t \gg 1/k$), peak amplitude/time at known locations, or drug-blockade conditions – rather than more point-wise observations of the same kind.

Sensitivity to sparse and noisy sampling. The noise sweep (Section 3.3) shows recovery remains within the 15% envelope at $\eta = 2\%$ (within the FSCV instrument-noise range [21]), deteriorates sharply at $\eta = 5\%$, and collapses at $\eta = 10\%$. The parameter-recovery grid (Section 3.5) adds a second aspect: when $1/k^* > T$ (fewer than one decay e-folding in the observation window), the data carries weak information about k and its recovery error grows substantially.

Empirical signature of the coupling. The trade-ray structure is directly visible in the multi-seed ensemble (Section 3.2): across all five seeds, D and k are simultaneously biased *above* truth ($9.9\% \pm 1.1\%$ and $14.6\% \pm 1.1\%$) in matched directions, with inter-seed SD under 1.1% – a systematic feature of the loss landscape, not optimizer drift. The Bayesian analysis (Section 3.4) corroborates

this: the Laplace posterior collapses k to the boundary $k \rightarrow 0$ while leaving D well-constrained, mirroring the unphysical $k_{\text{LM}} = -0.04$ from the Levenberg–Marquardt baseline (Section 3.6) – two independent manifestations of the same identifiability limit.

Implications for biological interpretation. Three implications follow. First, recovered (D, k) should be interpreted *jointly*, since the trade ray makes either marginal estimate less reliable than the joint one. Second, comparisons across conditions (e.g. healthy vs. Parkinsonian signalling) should examine the joint direction of change along the trade ray rather than marginal D or k alone. Third, experimental protocols that break the trade ray – extended observation windows, paired iontophoretic release, or simultaneous receptor-binding measurements – are the most effective way to improve recovery beyond the bias floor characterised here, and should inform measurement-protocol design in any prospective application to real FSCV data.

4.4 Limitations and Scope

We explicitly delineate the scope and the limitations of the present study.

Reduced spatial dimensionality. The model is restricted to a planar 2D domain and a single-pulse release event; real perisynaptic geometry is fully 3D with structural heterogeneity from dendritic spines, astroglial processes, and tortuous cellular packing that this abstraction does not capture. The 2D restriction follows prior modeling literature for thin synaptic clefts [25] and keeps computational cost tractable; extension to 3D is a straightforward modification of the input layer and PDE residual, but whether the accuracy reported here persists in the heterogeneous 3D regime remains to be demonstrated.

Synthetic observation data and absence of FSCV validation. The inverse problem is validated only against synthetic noisy observations from a finite-difference reference solver of the same governing PDE used in training, which makes recovery easier than from genuine experimental data

subject to instrument response, recording artefacts, and unmodeled biological variability. *No validation against real fast-scan cyclic voltammetry recordings has been performed*, and we make no claim that the accuracy reported here transfers to in vivo data. Such validation, on FSCV data from healthy and Parkinsonian subjects [21], is a required next step before the recovered parameters can be interpreted as physiologically meaningful estimates of disease state (Section 4.5).

Linearised reuptake kinetics. Reuptake is modeled as first-order with rate k , whereas DAT-mediated transport is more accurately Michaelis–Menten with concentration-dependent saturation [5, 25]. The linearization holds when concentration is well below $K_m \approx 0.2 \mu\text{M}$, but immediately after a quantal release event local concentration may briefly exceed K_m by an order of magnitude, where the linear model can underestimate clearance. Extending to a Michaelis–Menten sink is methodologically straightforward but adds a third inverse-problem parameter that would aggravate the identifiability issue discussed above; we leave this to follow-up work.

Homogeneous medium and absent receptor kinetics. D is treated as a single effective scalar, whereas real perisynaptic medium is heterogeneous on the scale of cells and extracellular-matrix components [5, 17]; applying the same framework to a spatially-varying $D(x, y)$ field is a natural but identifiability-aggravating extension. The model also excludes binding to postsynaptic D1/D2 receptors, consistent with how receptor binding is typically modelled separately from extracellular diffusion in the dopamine literature [20], but is required for any quantitative comparison with receptor-level pharmacological measurements.

Single operating point. Results are reported at a single canonical hyperparameter setting. Run-to-run variability across seeds is small (Table 3, $\sigma_D/D < 1\%$, $\sigma_k/k < 1\%$), but a systematic sweep over collocation density, network depth, and adaptive loss-weight schemes [24] would further strengthen the statistical robustness of the recovered estimates.

4.5 Future Directions

These limitations delineate the most immediate paths for future work. Extending to fully 3D synaptic geometries reconstructed from electron microscopy would test whether the 2D forward accuracy reported here persists in the heterogeneous 3D regime. Replacing synthetic data with real FSCV measurements [21] is the single most important next experiment and a prerequisite for any clinical interpretation of the recovered parameters; the inverse PINN’s data-efficiency suits the sparse-observation regime of in vivo recordings, but it has not yet been demonstrated on such data, and we make no performance claims there until that work is done. Beyond the synaptic scale, coupling this local model with network-level reaction-diffusion descriptions of alpha-synuclein spread and dopaminergic neuron loss [4, 8, 18] would, longer term, support a multi-scale pipeline relating molecular-scale signalling deficits to the spatiotemporal patterns of clinical decline.

5 Conclusion

In this work, we presented and validated a physics-informed neural network framework for the forward and inverse problems of dopamine diffusion in the synaptic cleft, governed by a two-dimensional linear reaction-diffusion equation. The forward PINN attains an L2 relative error of 3.75% against the analytical solution at early times and agrees with the finite-difference reference solver across the full simulation window (full-window L2 of 1.52% against the FD reference), while the inverse formulation simultaneously recovers the diffusion coefficient D and the reuptake rate k with mean relative errors of $9.85\% \pm 1.07\%$ and $14.61\% \pm 1.13\%$ (averaged across a five-seed network-initialization ensemble) at 2% observational noise on synthetic data. These results establish PINNs as a mesh-free, data-efficient alternative to classical finite-difference and finite-element solvers for quantifying synaptic neurotransmitter dynamics, with the inverse formulation in particular offering a tractable route to subject-specific parameter estimation that avoids the costly nested-optimization loops required by mesh-based pipelines [25]. Future work will extend the framework to fully three-dimensional synaptic geometries reconstructed from electron microscopy, incorpo-

rate measurements from fast-scan cyclic voltammetry recordings of striatal dopamine [21], and couple this local synaptic model with the network-level reaction-diffusion descriptions of Parkinson's disease progression [4, 8, 18] to construct a multi-scale computational pipeline linking molecular signaling deficits to clinical decline.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors have declared that no competing interests exist.

Ethics approval

Not applicable. This study did not involve human participants, human data, or animals; all reported results are derived from synthetic observations generated by a finite-difference reference solver.

Consent to participate / Consent to publish

Not applicable.

Data availability

All data underlying the results presented in this study are publicly available. The synthetic datasets used for forward- and inverse-problem validation, the trained PINN model configurations, and analysis notebooks are deposited in a public GitHub repository (<https://github.com/Hachero98/dopamine-PINN->) and archived on Zenodo with the permanent digital object identifier [10.5281/zen-](https://doi.org/10.5281/zenodo.105281)

[odo.20352595](https://doi.org/10.26434/chemrxiv-2023-odo.20352595). A complete reference run including all generated figures and the `metrics.json` produced by the reproducer pipeline is included in both repositories.

Code availability

All Python (JAX/Flax NNX) code used to produce the results reported in this manuscript is released under the MIT License in the same repository referenced in the Data availability statement. The repository includes a README with reproduction instructions, a pinned `requirements.txt`, and seed-controlled scripts and a Colab notebook that regenerate every figure and table in the paper.

Author contributions

Conceptualization: E.H., H.Z.; Methodology: E.H.; Software: E.H.; Formal analysis: E.H.; Investigation: E.H.; Writing – original draft: E.H.; Writing – review and editing: E.H., H.Z.; Supervision: H.Z. Both authors read and approved the final manuscript. **[EMMANUEL: PROPOSED split based on the standard advisor/student pattern – Emmanuel, confirm this is accurate with Huiqing before submission and adjust if her role included more than supervision and review (e.g. methodology input, funding acquisition).]**

Artificial intelligence disclosure

The authors used a large language model as an editing assistant during the preparation of this manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. No artificial intelligence was used to generate, fabricate, or interpret scientific data, results, or conclusions.

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